MATTHEW YU

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EDUCATION

University of Waterloo September 2025 - June 2030
Bachelor of Applied Science (B.A.Sc.) in Computer Engineering *Waterloo, ON*

EXPERIENCE

Statistics Canada — Software / Data Engineering September - Present

- Migrating **legacy SAS workflows** to open-source **Python and R** for improved portability and maintainability
- Building **reproducible data pipelines** for automated statistical processing, validation, and standardized outputs
- Developing a **GUI** to make modernized statistical workflows accessible and configurable for end users

HomeWise AI — Software Engineer Intern January 2026 - April 2026

- Built and shipped an IoT monitoring platform supporting **9 pilot properties**, enabling real-time telemetry ingestion and automated risk assessment
- Designed AWS data pipelines integrating REST APIs and streaming **real-time sensor data** into BigQuery for operational analytics
- Developed **PyTorch** and **XGBoost** models with time-series features, achieving $\mathbf{R^2 = 0.871}$ on held-out validation data

J Healthcare Initiative — Data Science / ML Research July 2026 - Present

- Developed **Python-based predictive models** to estimate how harm-reduction strategies affect **4 overdose-related outcomes**
- Developed **Negative Binomial and Poisson GLMs** on **90+ months** of drug-checking and health data
- Ran **1,000+ Monte Carlo simulations** to quantify uncertainty in projected policy outcomes

PROJECTS

Agentic Ecommerce Platform | ***Python, LangGraph, Supabase, Stripe, APIs*** April 2026

- Built a **$5K+ revenue** e-commerce platform with **Next.js**, **Stripe**, **Supabase**, and marketplace APIs
- Developed a **LangGraph agent** for order ingestion, SKU mapping, supplier routing, and fulfillment
- Integrated **LLM tool-calling**, human-in-the-loop review, guardrails, and audit logs for reliable order processing

AlphaZero-Style Othello Agent | ***Python, NumPy, PyTorch, Matplotlib*** March 2026

- Built a deep reinforcement learning agent to play Othello using **self-play**, **MCTS**, and a residual convolutional policy-value network in **PyTorch**
- Optimized Python ML prototype performance with **Numba** `@njit` JIT compilation and multiprocessing CPU workers with **CUDA** inference, increasing self-play data generation speed by over **60×**
- Evaluated models with custom metrics and **Matplotlib** visualizations to guide targeted rollbacks

TECHNICAL SKILLS

Languages	Python, C++, R, SQL, SAS
Frameworks	FastAPI, PyTorch, XGBoost, scikit-learn, NumPy, pandas, R Shiny
Infrastructure	AWS (Lambda, ECS, S3, CloudWatch, EventBridge), Docker, BigQuery
Tools	MS Excel, Git/GitHub, Postman, Airtable, REST APIs, Vercel